UTHEALTH HOUSTON AND MEMORIAL HERMANN HIGH RISK LIVER TRANSPLANT PROTOCOL FOR HEPATOCELLULAR CARCINOMA V7

Expanded tumor burden, treated vascular invasion and prior immune checkpoint exposure require different assessments. Use this protocol to classify the patient, direct treatment and reassessment, and determine whether the next action is standard-pathway care, further evaluation, a defined study or no transplant.

Version 7 | Revised 2026-10-24 06:55 CDT | Institutional approval pending

POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

1 PURPOSE AND SCOPE

Patients who satisfy current standard HCC criteria should remain on the standard pathway. Prior immune checkpoint inhibitor (ICI) exposure does not by itself preclude an HCC exception. Planned investigational ICI downstaging, burden beyond standard downstaging limits and treated portal-vein tumor thrombus (PVTT) must not be merged into a single permissive expanded category. [1] [2] [3]

A transplanting expansion requires a defined entry population and prospective oversight. Screening or tumor response is not permission to transplant. Preserve all pretreatment measurements after response; a favorable score predicts prognosis within its derivation population, not treatment benefit or allocation entitlement. [4] [5]

2 MULTIDISCIPLINARY TEAM

- Transplant surgery and hepatology coordinate decisions with HPB surgery, medical oncology, interventional and diagnostic radiology, radiation oncology, pathology, transplant pharmacy, nutrition and psychosocial services.
- For a study, add the research coordinator, biostatistician and independent safety board. For prior ICI, document hepatology, oncology and pharmacy review of immune toxicity and graft risk. Donor assessment remains independent.

3 PATIENT SELECTION AND BASELINE WORKUP

3.1 CONFIRM DIAGNOSIS AND MAP THE ORIGINAL DISEASE

- Order or obtain high-quality multiphasic liver CT or MRI, chest CT, AFP, CBC, bilirubin, albumin, INR, creatinine and sodium. Document Child-Pugh class, calculated MELD, ECOG performance status, portal hypertension, frailty and surgical fitness. [3] [6]
- Radiology must list each lesion with baseline segment, diameter and diagnostic category; distinguish viable tumor from treated nonviable tissue. Resolve atypical or discordant lesions through repeat appropriate imaging or MDT-directed biopsy when the result will change management. Do not count an indeterminate or LR-4 lesion as definite HCC solely to meet a transplant threshold. [3] [1]
- Obtain a written resection/ablation assessment and classify vascular findings as bland thrombus, tumor thrombus or unresolved. PET is not a substitute for liver imaging or a mandatory negative test for all HCC candidates. [3]

Advance only when the diagnosis, baseline burden, liver reserve, absence or presence of extrahepatic disease, and curative alternatives have been adjudicated. Hold for uncertain vascular invasion, discordant imaging or unexplained marker rise. Confirmed extrahepatic metastases or uncontrolled progression stops this proposed transplant route. [2]

3.2 EXPANDED BURDEN AND BIOMARKERS

- Record whether baseline disease met Policy 9.5.I downstaging entry criteria. Beyond-UNOS downstaging requires an individually justified nonstandard request if exception priority is sought; reaching T2 does not retrospectively confer standardized entry eligibility. [1] [2]
- Track AFP as a dated series. For standardized eligibility, a history of AFP above 1000 ng/mL requires treatment response below 500 ng/mL and maintenance below that level. Do not substitute a percentage fall for the applicable threshold. A guidance-ineligible AFP course requires separate review, not automatic enrollment. [1] [2]

XXL enrolled 74 patients and randomized 45 after downstaging selection; five-year OS from randomization was 77.5% with LT versus 31.2% with control treatment. The main report excludes macrovascular invasion, but the protocol appendix allowed uncertain bland-versus-neoplastic segmental/type-I portal thrombosis under additional response conditions. No separately reported proven-PVTT subgroup establishes efficacy for that phenotype. In the prospective MERITS-LT cohort, 76/190 beyond-UNOS candidates and 155/326 UNOS-DS candidates reached LT. Five-year entry-pathway OS was 45.0% versus 58.8%, despite recipient OS of approximately 72% versus 74%; recipient recurrence was approximately 30% versus 14%. These are different denominators, and five-year estimates require their small late risk sets and 27-month median follow-up to be considered. [5] [4]

HCC-B1 — HOW FAR BEYOND UNOS-DS SHOULD THE PROGRAM ATTEMPT DOWNSTAGING?

QUESTION AND COMPARISON

Adults with initially unresectable HCC outside UNOS downstaging entry limits, without extrahepatic disease or viable macrovascular invasion; compare a bounded downstaging-to-transplant strategy with standard UNOS-DS and nontransplant management. Evaluate treatment-entry survival, dropout, probability of transplant, recipient survival and recurrence separately.

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Yao 2015 [7] Single-center prospective downstaging protocol; 118 downstaging patients versus 488 initially T2 controls, treated in 2002–January 2012. Primary source | TMC: retrieve PDF Locator: Primary PMC full text: Methods, Results and survival figures; DOI 10.1002/hep.27752. Complete PDF not saved. Review depth: Primary PMC full text, Methods/Results and outcome figures reviewed; complete PDF not saved.	**Reported result:** Downstaging succeeded in 77/118 (65.3%); 64/118 (54.2%) underwent LT. Five-year post-LT survival was 77.8% versus 81.0%; recurrence-free probability 90.8% versus 88.0%. Five-year intention-to-treat survival was 56.1% versus 63.3%; two-year dropout 34.2% versus 25.6%. Five transplanted downstaging patients developed recurrence. **Do not infer:** Bounded protocol, older treatments and selected recipients; recipient survival excludes downstaging failure. Recurrence-free probability is not automatically a death-inclusive DFS endpoint.
Mehta 2021 [8] Prospective MERITS-LT cohort: 209 UNOS-DS patients at seven centers in four US regions, 2016–2019. Primary source | Full-text PDF Locator: Complete local author manuscript, PDF pp6–9, Table 3. Review depth: Complete local author manuscript reviewed, including outcome tables.	**Reported result:** 174/209 (83.3%) achieved downstaging; two-year estimated success 87.7%. Observed dropout was 75/209, including 51 tumor progressions and nine liver-related deaths; two-year competing-risk dropout 37.3%. LT occurred in 63/209 by cutoff; three-year LT probability 46.6%. Median post-LT follow-up 1.7 years; two-year recipient survival 95%, three-year 83.1%; recurrence 5/63. Three-year treatment-entry survival 73%. Explants exceeded Milan in 42.8%, with microvascular invasion in 17.5%. **Do not infer:** Early post-LT follow-up and substantial radiology/explant discordance; success of an imaging endpoint does not establish absence of viable cancer. Overlaps later MERITS expansions.
Sinha 2019 [9] Single-center all-comers analysis: 74 beyond-UCSF-DS patients versus 133 bounded UCSF-DS patients. Primary source | TMC: retrieve PDF Locator: Published primary abstract, PMID 30 779 440, Results. Review depth: Published primary abstract only; complete full text needed.	**Reported result:** Downstaging success: 64.8% versus 84.2%. Three-year dropout: 80.0% versus 36.1%. LT: 10/74 (13.5%) all-comers versus 59.0% of bounded patients. Five-year intention-to-treat survival: 21.1% versus 56.0%; three of ten all-comer recipients recurred. Number plus largest diameter >8 and Child-Pugh B/C predicted dropout. **Do not infer:** Strong counterevidence to unlimited entry, but small historical single-center cohort. Primary abstract only; treatment-entry timing and endpoint definitions require full text. Not a validation of a ≤12 cutoff.

Study • population • source	Results • interpretation boundary
Natarajan 2023 [10] Prospective MERITS-LT expansion: 82 all-comers versus 229 UNOS-DS patients, seven centers, 2015–2020. Primary source | TMC: retrieve PDF Locator: Primary PMC full text, Results §3.5 and outcome tables, PMC10 998 692; PDF pending. Review depth: Primary PMC Methods/Results and outcome tables reviewed; complete PDF not saved.	**Reported result:** Twelve-month downstaging: 67% versus 83%. Three-year entry survival: 58% versus 69%; approximately 30% in all-comers with Child-Pugh B/C or AFP ≥500. LT occurred in 20/82 versus 85/229; five-year LT probability 42.2% versus 73.6%. Recipient three-year survival: 90.9% versus 80.6%; recurrence 1/20 versus 7/85. Median post-LT follow-up was only about 18.5 months. All-comer dropouts numbered 39: 25 before and 14 after downstaging. **Do not infer:** Immature recipient estimates and substantial attrition. Cohort overlaps earlier MERITS and Xu 2025; do not sum patients or treat as independent validation.
Xu 2025 · all-comers [4] National prospective cohort, nine centers, 2015–2023: 190 all-comers and 326 UNOS-DS patients; intention-to-treat clock starts at first locoregional treatment. Primary source | Full-text PDF Locator: Complete local PDF, Results pp6–9, Tables 3–4; proposed refinements in Discussion pp9–11. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** Two-year downstaging: 66% versus 82%. Among all-comers, number plus largest diameter <10, 10–12 and >12 had success of 72%, 51% and 39%. Five-year treatment-entry survival: 45.0% versus 58.8%. LT occurred in 76/190 versus 155/326; median downstaging-to-LT 10.6 months. Five-year recipient survival: 71.5% versus 73.6%; recurrence: 29.8% versus 13.7%, with median post-LT follow-up 27.4 months. AFP ≥100 and Child-Pugh B predicted worse entry survival. **Do not infer:** Selected-recipient success cannot replace entry survival; five-year tails are immature. Y90/TACE comparisons were observational. The proposed number-plus-diameter ≤12, AFP <100 and Child-Pugh A combination remains unvalidated.
XXL 2020 [5] Multicenter randomized phase 2b/3 trial: 74 entered downstaging; 29 dropped out before randomization; 45 responders were randomized, 23 to LT and 22 to nontransplant treatment. Primary source | Full-text PDF Locator: Complete accepted manuscript and protocol appendix; final published abstract for survival estimates. Avoid mixing accepted-manuscript and final-publication hazard ratios. Review depth: Complete accepted manuscript including protocol appendix plus final published abstract; version-dependent hazard ratios not merged.	**Reported result:** From randomization, five-year overall survival was 77.5% with LT versus 31.2% without LT; tumor-event-free survival 76.8% versus 18.3%; median follow-up 71 months. **Do not infer:** Strongest causal evidence for LT among selected responders, not for all 74 entrants or unlimited initial burden. Small, prematurely stopped trial. Appendix allowed uncertain bland-versus-neoplastic segmental/type-I portal thrombosis under additional response conditions; the actual PVTT denominator/outcomes are not reported. It must not be relabeled a proven-PVTT trial.

Study • population • source	Results • interpretation boundary
Huang 2023 [11] UNOS national cohort, 2010–2019: 23,398 HCC exception applicants—20,579 initially Milan, 2,151 UNOS-DS, 668 all-comers. Primary source | TMC: retrieve PDF Locator: Primary abstract plus publisher Methods/Results, DOI 10.1016/j.cgh.2022.08.023; complete PDF pending. Review depth: Primary abstract and publisher Methods/Results reviewed; complete PDF not saved.	**Reported result:** Two-year dropout: 19.3%, 25.3% and 29.7%, respectively; the direct all-comer versus UNOS-DS difference was not significant (P=.07). In all-comers with initial AFP >100, two-year dropout was 52%, versus 26% in the comparison beyond-Milan patients. Higher MELD and total tumor diameter predicted dropout. **Do not infer:** Useful real-world waitlist counterevidence, but excludes patients lost before exception application and cannot supply first-LRT intention-to-treat survival. National registry overlaps other US registry studies.
Tabrizian 2022 [12] Five-center US recipient cohort, 2001–2015: 2,645 transplanted patients, including 2,122 always within Milan, 341 downstaged and 182 still beyond Milan. Primary source | TMC: retrieve PDF Locator: Primary PMC Methods/Results and survival tables, PMC9 301 590; complete PDF pending. Review depth: Primary PMC Methods/Results and survival tables reviewed; complete PDF not saved.	**Reported result:** Ten-year post-LT overall survival was 61.5%, 52.1% and 43.3%; ten-year recurrence was 13.3%, 20.6% and 41.1%, respectively. Median follow-up was 55.3 months. **Do not infer:** Longer-term cancer risk persists after successful downstaging. Recipient-only retrospective data do not estimate entry survival or probability of ever reaching LT; long-term tails have fewer patients.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Evidence supports offering a bounded downstaging pathway to selected beyond-UNOS patients, not unrestricted transplantation. Attrition before LT materially changes the counseling estimate: survival among recipients is consistently better than survival from treatment entry. Randomized XXL supports LT after sustained response but does not validate the proposed baseline burden/AFP/reserve combination. The ≤12, AFP <100, Child-Pugh A pilot envelope is a deliberately conservative candidate rule derived from observational risk signals. Observation is a test of behavior, not proof that dormant metastases are absent. [7] [8] [9] [10] [4] [5] [11] [12]

- Favorable MERITS recipient survival and the randomized responder benefit argue against declaring all beyond-UNOS disease futile. Conversely, high dropout in historical all-comers, worse entry survival, increased long-term recurrence and explant understaging argue against accepting a radiographic Milan endpoint alone. Child-Pugh B/C patients may need transplantation most, yet poorer downstaging tolerance makes an initial Child-Pugh-A-only pilot a local safety restriction rather than a universal oncologic exclusion.

Certainty / applicability: Moderate for a benefit of LT in carefully selected sustained responders, limited by one small responder-randomized trial; low for the exact expanded-entry limits; very low for a jointly validated ≤12/AFP <100/Child-Pugh A/six-month rule. This is an informal certainty assessment, not a formal GRADE exercise.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Number-plus-largest-diameter ≤12 and AFP <100/Child-Pugh A: cohort-derived risk refinement, not validated cutoff. Six months: proposed conservative operational observation period; studies used differing clocks and intervals. Imaging/AFP every three months: proposed monitoring cadence. A five-year ≥60% intention-to-treat target is a research-design benchmark, not an achieved or validated eligibility estimate for this pilot. The clinical stability clock must remain separate from OPTN/NLRB exception timing.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Xu 2025 · all-comers [4] "sum of tumor number and largest tumor diameter ≤12" Author-manuscript PDF p11, conclusion Source | Full-text PDF	Observationally proposed upper burden boundary, not an externally validated treatment-benefit rule.
Xu 2025 · all-comers [4] "exclusion criteria of AFP ≥100 and Child Pugh B/C" Author-manuscript PDF p11, conclusion Source | Full-text PDF	Supports a narrower initial cohort; does not establish that all excluded patients cannot benefit.
AASLD HCC 2023 [3] "3-to-6-month period of observation" Author-manuscript PDF p56, guidance statement 36 Source | Full-text PDF	Six months selects the conservative end of this guidance range; it is not a guarantee of favorable biology.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Open a bounded prospective beyond-UNOS downstaging cohort, not an unrestricted all-comers pathway. [4] [3] [5] [13] [7] [8] [9] [10] [11] [12]

- Proposed entry ceiling: original pretreatment tumor count plus largest tumor diameter in cm ≤12; AFP <100 ng/mL; Child-Pugh A; no macrovascular invasion, extrahepatic disease or unmeasurable infiltrative tumor.
  - These restrictions apply to the expanded arm, not to otherwise standard-eligible HCC.
- Before LT: viable tumor burden within Milan, or complete nonviability, with at least six months of documented stability from the first assessment meeting that target.
  - Repeat liver CT/MRI and AFP at least every three months; confirmed progression holds the route and resets the study control clock after new successful downstaging.
- Preserve baseline measurements and AFP.
  - A prior AFP ≥100 or Child-Pugh B/C at expanded-arm entry belongs to a separately approved stratum; do not erase that history after response.

– No initial stacking with the PVTT arm.

Why retain or change it: OPEN Analyst A position: retain the bounded prospective pilot, with original number plus largest diameter ≤12, AFP <100 and Child-Pugh A as initial enrollment restrictions; no extrahepatic disease, viable macrovascular invasion or infiltrative pattern. Require a documented within-Milan viable burden or complete radiographic nonviability and at least six months of stability before elective LT review. Preserve original burden and AFP rather than rewriting history after response. Prior AFP ≥100 or Child-Pugh B/C should enter a separately approved stratum, not be described as universally ineligible. A radiographic complete response is not explant proof of sterilization.

INFORMATION REQUIRED BEFORE THE DECISION

Central review of pretreatment and current multiphasic CT/MRI; chest CT; dated AFP series; Child-Pugh components; resection/ablation assessment; dated response and progression record. Record first treatment, first successful downstaging, latest progression, listing, first exception request and LT dates separately; report all-entry, waitlist and recipient outcomes with their own time origins.

Clinical versus allocation: Reaching Milan/T2 does not retrospectively create standardized downstaging entry eligibility. A nonstandard exception request, calculated-MELD route or LDLT review remains separate; apply the actual allocation clock.

Roadmap / protocol reconciliation: Retain the HCC roadmap's resection/ablation assessment and treatment selection. This proposal changes transplant entry boundaries, not the standard cancer-treatment algorithm.

Meeting record: HCC-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

3.3 PRIOR PORTAL VEIN TUMOR THROMBUS

- Record the actual involved vessel and original Vp grade: Vp1 distal to second-order branches; Vp2 second-order; Vp3 first-order right or left portal branch; Vp4 main trunk or contralateral branch. Record hepatic-vein involvement separately. Do not equate Vp3 with main-portal invasion or bland thrombosis with cancer invasion.

- Assess parenchymal and vascular response separately on serial contrast imaging, with expert radiology adjudication. Current NLRB guidance identifies main portal or hepatic-vein invasion as an HCC-exception contraindication; some treated primary-branch invasion or rupture cases may be considered only after at least 12 months of stability. A six-month study interval cannot replace this allocation interval. [2]

Yoon et al. reported 137 selected PVTT LDLT recipients, not all assessed PVTT patients. Five-year OS was 66.6% in 83 radiographic responders versus 35.1% in 54 nonresponders. Sixty-five of 83 responders still had viable PVTT on explant. Only 14 recipients had Vp4 disease. This recipient-only, nonrandomized experience cannot establish a routine Vp4 pathway, a causal LT benefit, or equivalence between vascular grades. [14]

HCC-B2 — WHICH PVTT ANATOMY COULD ENTER A TRANSPLANTING STUDY?

QUESTION AND COMPARISON

HCC with previously documented portal-vein tumor thrombus, after antitumor treatment; compare original segmental/second-order versus first-order/main-trunk extent and response strata. Outcomes are treatment-entry access and survival, recipient survival, recurrence and graft-related harm—not anatomy alone.

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Yoon 2026 [14] Retrospective Asan LDLT cohort, 1997–2022, 137 selected PVTT recipients; median follow-up 38.3 months. Primary source | Full-text PDF Locator: Complete PDF: Table 1; Results p7261; Table 2 p7264; Figure 1 p7265; Discussion p7268. Preserve source/pvtt_outcomes.py values. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** Vp1 five-year OS 69.5%, DFS 56.2%; Vp4 OS 50.0%, DFS 34.3%. Exact Vp2/Vp3 five-year values were not tabulated. Vp4 recurrence was 9/14 (64.3%) by Table 2 count, not a five-year risk. Responders: 83 patients, five-year OS 66.6%, DFS 64.7%, recurrence 28/83; nonresponders: 54, OS 35.1%, DFS 19.7%, recurrence 42/54. Viable PVTT remained in 65/83 responder explants. **Do not infer:** Recipient-only, heterogeneous treatment eras and small grade subgroups. Nonsignificant grade comparisons do not prove equivalence. Discussion/conclusion transpose responder OS and DFS; retain Results/Figure 1 values.
Choi 2017 [15] Retrospective selected LDLT recipients with PVTT, n=34. Primary source | TMC: retrieve PDF Locator: Primary published abstract, PMID 28 480 517; full text required for endpoint definitions and patient selection. Review depth: Primary published abstract only; complete full text needed.	**Reported result:** Five-year OS was 50.3% for segmental and 14.3% for first-order/lobar involvement. Segmental five-year DFS was 63.9%; a numeric lobar DFS estimate was not available in the verified primary abstract. **Do not infer:** Very small lobar group, retrospective recipient selection and no all-downstaging-entry denominator. OS and DFS are retained as reported, not converted into recurrence risk.
Yang 2020 [16] Retrospective 75-recipient PVTT cohort. Primary source | TMC: retrieve PDF Locator: Primary journal abstract, PMID 32 303 439, Results. Review depth: Primary journal abstract only; complete full text needed.	**Reported result:** Three-year OS/RFS were 65.4%/40.0% for Vp2–3 combined and 30.6%/21.4% for Vp4. **Do not infer:** Do not split combined Vp2–3 results into invented grade-specific estimates. No verified all-entry denominator or grade-specific recurrence counts; endpoint death handling needs full text.

Study • population • source	Results • interpretation boundary
Ye 2024 [17] Retrospective DDLT cohort of 141 vascular-invasion recipients: 13 segmental PVTT, 68 lobar PVTT, 60 microvascular invasion. Primary source | Full-text PDF Locator: Complete 12-page PDF: Methods p3, Tables 1–2 p6, Figure 4 p7. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** Segmental three-year OS 75.0%, RFS 58.3%, crude recurrence 4/13. Lobar three-year OS 45.9%, RFS 9.1%, crude recurrence 60/68. **Do not infer:** Exclusion of early deaths biases recipient survival upward. Lobar is not an isolated Vp3 group. Crude recurrence and RFS are distinct measures; the cohort is not directly transportable to Western mixed-etiology candidates.
Soin 2020 [18] Cohort following a PVTT downstaging protocol: 43 treated, 27 responders, 25 LDLT recipients. Primary source | TMC: retrieve PDF Locator: Primary publisher HTML Methods/Results and abstract, DOI 10.1097/TP.0000 000 000 003 162; complete PDF pending. Review depth: Primary abstract and publisher HTML Methods/Results reviewed; complete PDF not saved.	**Reported result:** Reported five-year OS 57% and RFS 51% exclude two postoperative deaths, leaving a 23-recipient survival subset. Recurrence occurred in 6/25 (24%) of the whole transplanted group. **Do not infer:** Neither reported survival curve represents all 25 recipients or all 43 treatment entrants. Small selected cohort without randomized control; grade-specific outcomes unavailable.
Assalino 2020 · vascular invasion [19] Multicenter retrospective series: 45 transplanted patients reviewed, 30 with centrally validated macrovascular invasion retained; 11 centers, 2004–2018. Primary source | Full-text PDF Locator: New complete publisher PDF pp569–573: Methods, Table 1, explant findings and limitations. Review depth: New complete 9-page publisher PDF, including Methods, anatomy definitions, results, tables and references.	**Reported result:** Median post-LT follow-up 39 months; one/three/five-year OS 76.7%/66.2%/59.6%; DFS 63.3%/56.3%/56.3%. Recurrence 8/30, median 6.5 months after LT. Three explants retained major vascular invasion despite radiologic clearance. An exploratory AFP <10 subgroup had 11% recurrence and five-year OS 83.3%. **Do not infer:** The 15 excluded records are not downstaging failures. No all-treated denominator. AFP threshold was post hoc; subgroup denominator not verified. Neither median waiting time nor the mixed Vp3 label validates a 12-month or modern Vp3 cutoff.

Study • population • source	Results • interpretation boundary
Sposito 2026 [20] Five-center Italian Y90 cohort, 2010–2023: 240 consecutive HCC/PVTT patients; 61 downstaged, 37 transplanted. Primary source | TMC: retrieve PDF Locator: Published primary abstract, PMID 42 228 945, DOI 10.1097/HEP.0000 000 000 001 801. Review depth: Published primary abstract only; complete full text needed.	**Reported result:** Downstaging: 61/240 (25.4%); LT: 37/240 (15.4%). Among downstaged patients, 60-month cancer death was 15.8% with LT versus 45.2% without LT (subdistribution HR .221, 95% CI .071–.683). Recipients had three/five-year OS 67.2%/61.6% and RFS 79.3%/79.3%; complete explant necrosis 56.7%. **Do not infer:** Abstract-only verification. Grade denominators, landmark/selection adjustment and RFS death handling require full text; RFS exceeding OS must not be converted to recurrence as 1–RFS. No verified all-240 survival estimate.

PVTT OUTCOMES UNDERLYING THE OPEN ANATOMY BOUNDARY

The evidence supports studying carefully selected patients, but does not validate a universal Vp2/Vp3 cutoff. The following are outcomes after transplantation in selected recipients, not the chance of benefit from the start of downstaging. OS means overall survival; DFS/RFS retain each paper's terminology. Observed recurrence is a crude count over available follow-up, not a fixed-time cumulative incidence.

INDIVIDUAL VP GRADES IN THE LARGEST LDLT COHORT REVIEWED

Yoon 2026: 137 recipients, Asan, 1997–2022; median follow-up 38.3 months. Record original highest Vp extent and current viable extent separately; do not re-label historical Vp4 as Vp1 after response. [14]

Vp grade and vessel	n	5-year OS	5-year DFS	Observed recurrence
Vp1 Peripheral to second-order branches	51	69.5%	56.2%	Not separately tabulated
Vp2 Second-order branches	36	Not tabulated	Not tabulated	Not separately tabulated
Vp3 First-order right or left branch	36	Not tabulated	Not tabulated	Not separately tabulated
Vp4 Main trunk and/or contralateral branch	14	50.0%	34.3%	9/14 (64.3%), counted from Table 2; not a five-year risk

Source locators: Table 1, Results p7261, Table 2 p7264 and Figure 1 p7265. The Vp2/Vp3 survival curves are shown, but exact five-year estimates are not supplied numerically; no values have been guessed from the graph. [14]

Yoon 2026 — Full-text PDF

Vp4 recurrence was independently counted from the 14 Table 2 rows. Grade-specific recurrence counts for Vp1–3 were not separately reported. Neither 100 minus DFS nor 100 minus RFS is relabeled as recurrence. Different event definitions, death handling and follow-up preclude that substitution.

ANATOMY COMPARISONS IN OTHER TRANSPLANT COHORTS

Study and reported group	OS	DFS or RFS	Observed recurrence
Choi 2017 Segmental, n=27; Vp1–2-type anatomy	50.3% at 5 y	DFS 63.9% at 5 y	Not in verified abstract
Choi 2017 First-order/lobar, n=7; Vp3-type anatomy	14.3% at 5 y	Numeric value not in verified abstract	Not in verified abstract
Yang 2020 Vp2–3 combined; 75 total recipients	65.4% at 3 y	RFS 40.0% at 3 y	Not in verified abstract
Yang 2020 Vp4; same 75-recipient cohort	30.6% at 3 y	RFS 21.4% at 3 y	Not in verified abstract
Ye 2024 Segmental, n=13	75.0% at 3 y	RFS 58.3% at 3 y	4/13 (30.8%)
Ye 2024 Lobar including main portal vein, n=68; NOT isolated Vp3	45.9% at 3 y	RFS 9.1% at 3 y	60/68 (88.2%)
Soin 2020 25 downstaged LDLT recipients; Vp1/2/3/4 = 1/12/11/1	57% at 5 y*	RFS 51% at 5 y*	24% (6/25); whole transplanted group

Choi and Yang rows are provisional primary-abstract extractions, not full-PDF review. Combined anatomical groups cannot supply separate Vp1, Vp2 or Vp3 estimates. [15] [16]

Choi 2017 — TMC PDF retrieval | Yang 2020 — TMC PDF retrieval

Ye: Table 2, PDF p6; Results and Figure 4, p7. Its “lobar” definition includes the main trunk (Methods p3), so these results cannot be assigned to Vp3 alone. Perioperative deaths and follow-up losses within one year were excluded; all patients had HBV. [17]

Ye 2024 — Full-text PDF

*Soin: the 57% OS and 51% RFS exclude two postoperative deaths. They are not outcomes of all 25 recipients or all 43 patients entering downstaging. Publisher HTML describes 27 successful downstagings and 25 transplants; complete PDF and supplementary tables remain pending. [18]

Soin 2020 — TMC PDF retrieval

RESPONSE MATTERS AND DOES NOT ERASE THE ANATOMICAL UNCERTAINTY

Yoon response stratum	n	5-year OS	5-year DFS	Observed recurrence
CR or PR by overall mRECIST	83	66.6%	64.7%	28/83 (33.7%)
Stable or progressive disease	54	35.1%	19.7%	42/54 (77.8%)

Yoon: Results p7261, Figure 1 and Table 4. CR/PR is overall mRECIST response, not mandatory complete disappearance of viable vascular tumor; 65/83 responders still had viable PVTT on explant. [14]

Yoon 2026 — Full-text PDF

INTERPRETATION FOR HCC B2

- The Yoon grade comparisons were not statistically significant (OS P=0.1882; DFS P=0.2988). With only 14 Vp4 recipients, this neither proves equivalence nor establishes futility. The Vp4 five-year OS of 50.0% is a real counterargument to blanket exclusion.
- Older anatomy-based cohorts report substantially poorer proximal-vessel outcomes; the populations, treatments, classification systems and endpoint definitions differ. Possible center/time overlap among Chinese cohorts prevents treating every series as independent confirmation. No pooled survival percentage is calculated.
- OPEN local position: retain original Vp1–2 as the initial transplanting-study boundary. Vp3 registry-first means observational follow-up without transplantation authorized by that registry; any transplanting Vp3 arm needs separate prospective approval. Keep Vp4 outside the initial pilot, not labeled universally futile.
- The complete vascular-response requirement and ≥12-month control period are conservative proposed safeguards, not thresholds validated by these cohorts. Favorable LDLT outcomes do not establish deceased-donor allocation priority or resolve living-donor risk.

Read the selection and reporting limits together with the outcome estimates. [14] [17] [15] [16] [18]

Yoon 2026 — Full-text PDF | Ye 2024 — Full-text PDF | Choi 2017 — TMC PDF retrieval | Yang 2020 — TMC PDF retrieval | Soin 2020 — TMC PDF retrieval

Evidence integrity note: Yoon's discussion/conclusion transpose the responder OS and DFS percentages; this update uses the Results and Figure 1 values. Request numerical Vp2/Vp3 Kaplan–Meier estimates, confidence intervals and grade-specific recurrence counts from the authors or a supplement before using exact grade-specific estimates in consent.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Anatomy, treatment response and biology all matter, but the available cohorts do not validate a sharp universal Vp2/Vp3 boundary or a single control-duration requirement. Several selected branch or even main-trunk recipients survived long term; this argues against a blanket futility claim. Conversely, small anatomy strata, highly selective transplantation, exclusion of perioperative deaths in some cohorts and absent entry denominators prevent a routine-policy expansion. Overall mRECIST response is not synonymous with vascular sterilization, and imaging disappearance can coexist with viable explant PVTT. [14] [15] [16] [17] [18] [19] [20]

- Poorer outcomes for lobar/main-trunk groups in Choi, Yang and Ye favor caution, while Yoon's selected Vp4 survivors and Sposito's downstaged cohort show that advanced anatomy is not invariably fatal. Assalino included hepatic-vein and mixed first-order/main-trunk disease, but its selected-recipient results cannot establish that these groups should enter an initial US pilot. Differences in anatomical definitions prevent a valid pooled Vp-grade survival estimate.

Certainty / applicability: Very low for a specific anatomy or waiting-time threshold; low for an association between response/biologic selection and better recipient outcomes. No reviewed randomized study directly established the transplant strategy's net benefit from confirmed-PVTT treatment entry.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Vp1–2 restriction, complete vascular radiographic response and 12-month control are proposed local safeguards, not a jointly tested published rule. Sposito accepted sustained vascular CR/PR for ≥6 months; Assalino reported a median, not a minimum, 11.4-month interval. NLRB guidance permits selected branch-invasion/rupture cases after prolonged stability but does not mandate clinical acceptance or convert main-trunk/hepatic-vein disease into a standard exception. Preserve separate original and current vascular extent.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Yoon 2026 [14] "Those with CR or PR were defined as DS responders" Published PDF p2 / journal p7260, response assessment Source | Full-text PDF	Yoon did not require the complete vascular clearance proposed here; 65/83 responders still had viable PVTT on explant.
Yoon 2026 [14] "our study did not identify it as an independent predictor of survival." Published PDF p10 / journal p7268, discussion of PVTT grade Source | Full-text PDF	The selected cohort did not establish a sharp anatomical survival cutoff. Vp4 five-year OS was 50.0%; small numbers and non-intention-to-treat selection prevent an equivalence claim.
Ye 2024 [17] "lobar PVTT: tumor thrombi involving left and right branches or main portal vein" Complete early-online PDF p3, Diagnosis of PVTT Source | Full-text PDF	This poor-outcome group combines first-order and main-portal involvement. Its recurrence cannot be attributed to Vp3 alone; perioperative deaths were excluded.
NLRB oncology guidance [2] "minimum of 12 months" December 2025 guidance p3, treated primary-branch invasion/rupture discussion Source | Full-text PDF	Applies to specified case-by-case exception considerations; applying this duration to every Vp1–2 study candidate is local conservatism.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Develop an initial Vp1–2-only transplanting study; keep Vp3 registry-first and Vp4/hepatic-vein expansion closed. [14] [17] [2] [15] [16] [18] [19] [20]

- Proposed anatomy: original Vp1 or Vp2 only, centrally confirmed.

  - Exclude Vp3, Vp4/main or contralateral branch, hepatic-vein tumor invasion, previous tumor rupture and combined expanded-burden risk from the initial pilot.
  - These are study boundaries, not universal declarations of futility.

- Require complete absence of enhancing viable vascular tumor, remaining viable parenchymal disease within Milan or completely nonviable, no extrahepatic disease, and ≥12 continuous months meeting these conditions.
  - Scan approximately every three months; count from the first qualifying scan.
- Any recurrent vascular enhancement or new metastasis stops progression to LT.
  - Indeterminate thrombus or imaging discordance is a hold, not automatic clearance.

Why retain or change it: OPEN Analyst A position: begin the transplanting study with original Vp1–2 only; collect original Vp3 in a registry without routine transplantation while the group reviews full-text evidence and a separately governed stratum. Keep original Vp4, hepatic-vein invasion, rupture and stacked high-risk features outside this initial pilot. Require no enhancing viable vascular component, controlled within-Milan parenchymal disease, no extrahepatic disease and ≥12 months documented control before elective review. This is a conservative institutional research boundary—not a validated futility boundary for excluded grades.

INFORMATION REQUIRED BEFORE THE DECISION

Original vessel-by-vessel map with Vp grade; expert serial contrast CT/MRI; chest staging; AFP; separate vascular and parenchymal response; resection options. Crosswalk the original study/anatomic vascular definition to current Vp grade. Record vascular response separately from overall mRECIST response; neither excludes microscopic viable PVTT.

Clinical versus allocation: Clinical study eligibility cannot override the main-portal/hepatic-vein exception contraindications or grant priority. Obtain a separate documented organ route.

Roadmap / protocol reconciliation: Continue standard HCC treatment and restaging while the study is developed; radiographic downstaging alone is not an instruction to list.

Meeting record: HCC-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

4 DOWNSTAGING TREATMENT AND RESPONSE ASSESSMENT

4.1 SELECT A LIVER DIRECTED TREATMENT PLAN

- Choose ablation, TACE, Y90 or external-beam treatment according to lesion distribution, vascular anatomy, liver reserve and treatment feasibility. Record why resection or a less burdensome definitive treatment is unsuitable. Observational modality differences do not justify a universal Y90 preference. [3] [4]
- For Y90, record microsphere type, dosimetry model, target volume and normal-liver exposure. For SBRT, record fractionation and organ-at-risk constraints. Do not use an isolated absorbed-dose number copied from a different device, compartment model or tumor population as an order.
- Proposed local monitoring: liver CT/MRI every 6–8 weeks during active downstaging, AFP every 4 weeks, and immediate assessment for decompensation or new symptoms. Adjust to modality and toxicity; record mRECIST response and original lesion sites. Imaging used for a beyond-standard downstaging exception should be at least four weeks after the last downstaging treatment. [2] [3]

4.2 ICI EXPOSURE AND SYSTEMIC THERAPY

- Use the HCC oncology roadmap to choose systemic treatment for the cancer indication. Adding an ICI or systemic agent solely to enable LT is not a routine transplant order. A transplant-specific plan must address residual graft risk, other drug washouts, treatment toxicity and continued cancer control. [3] [21] [22]

VITALITY reported 117 assessed patients, 65/86 initially beyond Milan downstaged, and 43/117 transplanted; most also received locoregional therapy. ImmunoXXL reported 16 selected transplanted patients from 61 treated, with four rejection events. These studies do not isolate an ICI effect from selection or concomitant therapy. Full-methods access limitations are recorded with the references. [21] [22]

- Record ICI agent, every exposure date, last dose, immune-related adverse events and treatment, active infection, renal function and other immunosuppression. The OPEN HCC-B3 proposal is an elective interval of at least 90 days plus a separate toxicity gate. It is a conservative local safeguard, not a proven safe minimum or an OPTN requirement. Shorter exceptional-care timing requires explicit independent review; longer intervals do not eliminate risk. [23] [24] [25]

In a 119-recipient international cohort there were 24 rejection events and six rejection-related graft losses. Shorter washout was associated with more rejection; the above-50-day reference group is not an evidentiary guarantee of safety. The 91-patient individual-patient meta-analysis is largely retrospective. Prior immune toxicity is a risk signal, not proof that ATG or higher tacrolimus targets prevents rejection. [23] [24] [25]

HCC-B3 — HOW SHOULD PRIOR ICI ALTER ACTIVATION AND CONSENT?

QUESTION AND COMPARISON

Patients with HCC treated with immune checkpoint inhibitors before planned LT; compare shorter versus longer ICI-free intervals, prior irAE versus none, and ICI-exposed versus non-ICI recipients. Assess treatment-entry tumor control/access to LT separately from recipient rejection, graft loss, recurrence and survival.

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
VITALITY · Tabrizian [21] Multicenter observational US cohort: 117 consecutive ICI-treated HCC patients assessed for LT, 2016–2023. Primary source | TMC: retrieve PDF Locator: Primary published abstract and accessible publisher material, PMID 39 255 928; full PDF pending. Review depth: Primary published abstract and accessible publisher material; complete full text needed.	**Reported result:** 65/86 (75.6%) beyond-Milan patients downstaged, median 5.6 months; 43/117 (36.7%) received LT. Three-year intention-to-treat survival was 71.1%; three-year recipient survival 85%. Rejection occurred in 7/43; six had last ICI <3 months before LT; one rejection-associated graft loss. **Do not infer:** The abstract's categories—18 initially within Milan plus 23 downstaged recipients—sum to 41, not 43; the two-patient discrepancy awaits full flow-diagram review. Multimodal treatment prevents attribution to ICI alone; full paper unavailable.

Study • population • source	Results • interpretation boundary
ImmunoXXL 2026 [22] Prospective phase II experience: 61 potentially eligible atezolizumab–bevacizumab-treated patients; 16 enrolled/listed/transplanted responders. Primary source | TMC: retrieve PDF Locator: Published abstract plus accessible publisher Results, PMID 41 763 363; full PDF needed for eligibility, flow and adverse-event denominators. Review depth: Primary abstract and accessible publisher Results sections; complete full text needed.	**Reported result:** Median washout 57.5 days (IQR 29–87), median follow-up 16 months. Acute rejection 4/16 (25%); three irAEs. Ninety-day morbidity 62.5%, mortality 1/16. Explants: ten complete and six partial responses; one recurrence. Two-year recipient OS 94%, RFS 90%. **Do not infer:** These are selected-16 outcomes, not 61-patient ITT efficacy. Small, nonrandomized, short follow-up; a shorter median washout does not establish safety. Retransplantation count was not independently extractable from accessible primary text and is not asserted.
Moeckli 2025 [23] International retrospective series of 119 ICI-exposed LT recipients. Primary source | TMC: retrieve PDF Locator: Primary abstract, PMID 40 042 053; complete publisher PDF inaccessible. Review depth: Primary published abstract; complete full text needed. Repository returned access challenge.	**Reported result:** Rejection occurred in 24/119 (20.2%), median nine days after LT. Six of 24 rejection cases lost the graft—6/119 (5.0%) of all recipients. Relative to >50 days, univariate rejection odds ratios were 21.3 (95% CI 5.93–103) for <30 days and 9.48 (2.47–46.8) for 30–50 days. **Do not infer:** Retrospective association, broad confidence intervals and selection by transplant timing; no untreated comparator or all-ICI-treatment denominator. >50 days is not a prospectively validated safe threshold.
Rezaee-Zavareh · IPD [24] Secondary systematic review and individual-patient-data analysis: 91 recipients from 30 predominantly case reports/series and cohorts. Primary source | Full-text PDF Locator: Complete local author manuscript: Results/washout model; PDF p10 (manuscript p7) conclusion. Existing V6 quotation ledger retained. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** 24/91 rejections (26.4%), nine recurrences and nine deaths; median follow-up 690 days. Adjusted rejection hazard ratio per additional washout week .92 (95% CI .86–.99). Ninety-four days was the median washout among modeled patients with predicted rejection probability ≤20%, not an observed universal safety cutoff. **Do not infer:** Publication/selection bias, heterogeneous regimens, small events and overlap with later cohorts. A modeled ≤20% probability does not justify telling an individual that three months eliminates rejection risk.

Study • population • source	Results • interpretation boundary
Fang · irAEs [25] Multicenter Chinese retrospective cohort, 2018–2024: 209 ICI-exposed LT recipients. Primary source | Full-text PDF Locator: Complete local PDF pp5–6, Tables 3–4; avoid reversed conditional denominators. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** Rejection occurred in 36/209 (17.2%); 61.1% biopsy proven, median ten days after LT. Correct risks from Table 3: 21/44 (47.7%) with prior irAE versus 15/165 (9.1%) without. Any irAE adjusted OR 9.17 (95% CI 4.00–21.02); washout <30 days OR 3.071 (1.211–7.787). The six grade 3–4 cases split three with and three without rejection. **Do not infer:** The discussion's 58.3%/13.3% are irAE prevalence within rejection/nonrejection groups, not rejection probabilities. Too few severe irAEs to validate a grade-specific exclusion or a recovery interval.
Aceituno 2026 [26] Ten-center ICI series, 48 recipients, propensity matched with 144 Toronto non-ICI controls; 20/48 exposed cases previously published. Primary source | Full-text PDF Locator: Complete local PDF pp2–5, Methods and Tables 1–2. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** Biopsy-proven rejection 9/48 (18.8%) versus 28/144 (19.4%), P=.916. Two exposed cases were steroid resistant; one rejection-related graft failure required retransplant. Overall graft failure 2/48 versus 3/144, retransplant 2 versus zero, recurrence 5/48 versus 15/144, death 7/48 versus 21/144. Median follow-up 2.6 versus 3.2 years. Table 1 washout median 60 days, IQR 22–196. **Do not infer:** No detected difference is not equivalence. Substantial tumor-risk imbalance and published-case overlap. Abstract washout IQR 13–96 conflicts with Table 1 and appears to match rejection-onset timing; use table and flag discrepancy. It cannot reassure about prior irAE risk.
LITCHI 2026 [27] Multinational European registry, ten centers, 18 recipients, 2018–2024. Primary source | TMC: retrieve PDF Locator: Primary full HTML and indexed publisher PDF Methods/Results, DOI 10.1111/liv.70692. Complete PDF download remains blocked. Review depth: Primary full HTML and indexed publisher PDF Methods/Results reviewed; actual complete PDF download blocked.	**Reported result:** Four of 18 developed biopsy-confirmed rejection (22%; 95% CI 6.4–47.6). One fatal rejection occurred 21 days after LT following 13-day washout; three other events responded to intensified immunosuppression. One HCC recurrence; one-year overall survival 77%. **Do not infer:** Small, noncomparative and largely long-washout cohort. Late rejection and incomplete protocol-biopsy surveillance preclude a guaranteed safe interval or extrapolation of the aggregate rate to short-washout patients.

Study • population • source	Results • interpretation boundary
HCC ICI cohort · JAMA 2026 [28] Eight-center Chinese recipient cohort, 2019–2024; initially beyond Milan; 208 ICI recipients propensity matched to 208 non-ICI recipients. Primary source | TMC: retrieve PDF Locator: Primary October 2026 JAMA Oncology abstract, PMID 42 821 267; full PDF pending. Review depth: Primary published abstract; complete full text needed.	**Reported result:** Median post-LT OS 4.57 versus 2.53 years, HR .63 (95% CI .43–.92); RFS 1.47 versus .89 years, HR .59 (.44–.80); recurrence subdistribution HR .54 (.40–.72). Ninety-day rejection was 34/208 (16.3%) versus 17/208 (8.2%). **Do not infer:** Favorable oncologic association coexists with greater rejection. Residual confounding and selection remain; there is no all-downstaging-entry ITT comparison. Possible center/year overlap with Fang and other Chinese cohorts must be checked before pooling. Abstract-only verification.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

ICI can help some selected patients reach transplantation, but efficacy and safety denominators must be kept separate: response among treated patients, probability of reaching LT, and rejection among recipients answer different questions. Very short washout is consistently concerning; evidence does not establish an interval at which risk becomes zero. Prior irAE adds information beyond elapsed time. Neither the apparent safety of some matched cohorts nor promising recipient survival cancels the graft-loss signal. [21] [22] [23] [24] [25] [26] [27] [28]

- ImmunoXXL demonstrates feasibility with intervals shorter than 90 days, and Aceituno found similar rejection proportions in matched groups; these challenge a claim that <90 days is an absolute biological contraindication. They do not prove equivalence or validate elective shortening. Ong suggests oncologic benefit but also higher rejection, while Fang highlights risk missed by washout alone. The literature supports individualized exceptional review, not automatic urgency-based exemption.

Certainty / applicability: Low for the association between shorter washout and rejection; low for prior-any-irAE risk; very low for a precise safe interval, a grade-specific irAE exclusion, an optimal immunosuppression regimen or an ICI-caused treatment-entry survival benefit. No reviewed trial randomized washout or transplant versus continued systemic therapy after ICI.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Ninety days: conservative local planning rule informed by retrospective timing associations and IPD modeling, not a validated threshold. Grade ≤1 recovery and exclusion of prior severe nonendocrine/myocarditis cases: proposed operational safety judgment, not a proven transplant-specific cutoff; Fang's severe-event sample is only six. No laboratory or immune assay identified in this review validates clearance for LT. Center-specific immunosuppression should not be portrayed as evidence-proven protection from ICI-associated rejection.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Rezaee-Zavareh · IPD [24] "a 3-month washout may reduce it" Accepted PDF p10 / manuscript p7, abstract conclusion Source | Full-text PDF	A risk-reduction suggestion from largely retrospective data, not a safe minimum.
Fang · irAEs [25] "irAEs (any grade vs non)" Published PDF p6, Table 4 Source | Full-text PDF	Prior irAEs were associated with rejection; the study did not validate the specific grade-based activation rule proposed here.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use ≥90 days after the last ICI dose as the elective LT planning threshold, with a separate immune-toxicity gate. [24] [25] [23] [21] [22] [26] [27] [28]

- Proposed elective cutoff: ≥90 days from last ICI dose to LT.
  - No interval is proven safe; a longer interval does not eliminate rejection or graft-loss risk.
- Require no active clinically significant non-endocrine immune toxicity: grade ≤1 and no escalating immunosuppression for toxicity.
  - Stable endocrine replacement may qualify.
  - Prior grade 3–4 non-endocrine toxicity or myocarditis is excluded from the initial pilot pending separate specialist review.
- A shorter interval has no automatic urgency waiver.
  - An exceptional-care decision must document near-term mortality without LT, alternatives, donor/graft risk, consent and rescue capability.
  - Biology, ICI and allocation clocks may overlap; do not add them mechanically.
- Consent must address rejection, graft loss, uncertain comparative oncologic benefit and progression during washout.
  - Neither elapsed time nor complete radiological response guarantees immune safety.
  - Obtain the missing primary full texts before ratifying these provisional thresholds.

Why retain or change it: OPEN Analyst A position: use ≥90 days since the last ICI dose for elective transplant planning, described as risk reduction rather than safety assurance. Before advancement, document the agent/dates, prior irAE organ and grade, treatment required, recovery and current immunosuppression; require clinically significant nonendocrine irAEs to resolve to grade ≤1 without escalating treatment. Stable endocrine replacement is not treated as active uncontrolled irAE. Prior grade 3–4 nonendocrine irAE or myocarditis remains outside the initial pilot pending specialist/committee review and a separately specified pathway. A shorter interval requires explicit exceptional review, not an automatic emergency waiver.

INFORMATION REQUIRED BEFORE THE DECISION

Drug and last-dose dates; complete irAE history and grade; treatment and recovery; graft-risk review by hepatology, oncology and pharmacy; current staging.

Clinical versus allocation: Prior ICI alone is not an automatic HCC exception exclusion. The proposed clinical timing gate is separate from tumor eligibility and allocation waiting time.

Roadmap / protocol reconciliation: Preserve effective cancer treatment choices, but discuss transplant intent before initiating ICI. No immune assay, ATG regimen or high tacrolimus target is validated as clearance.

Meeting record: HCC-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

4.3 ADVANCE OR HOLD AFTER TREATMENT

- Advance to selection review only with documented tumor control, no new extrahepatic disease, acceptable reserve and toxicity, repeat assessment of resection/ablation options, the applicable AFP response, and completion of the selected clinical and allocation observation requirements. [2] [3]
- Hold for unresolved lesions, marker/imaging discordance, incomplete ICI review or unapproved study boundaries. Stop the proposed route for uncontrolled progression, new metastatic disease or medical risk incompatible with LT. Do not continue increasingly toxic therapy simply to manufacture eligibility.
- Maintain separate dated clocks for response stability, ICI washout and exception waiting time. They may overlap when allowed; do not automatically add a new observation period after the ICI washout.

5 LISTING AND TRANSPLANT SURGERY

- Before activation, selection conference must sign the diagnosis and baseline map, final staging and AFP, tumor-control timeline, medical fitness, approved study stratum if applicable, actual allocation route and donor clearance. Any missing requirement keeps activation on hold. [1] [2]
- Before incision, reassess interval clinical or imaging changes and confirm graft/vascular technical planning. Unexpected disseminated malignancy stops transplantation under this pathway. Explant assessment records viable burden, differentiation, vascular invasion and treatment response.
- Before recipient activation, document the intended DDLT or LDLT route, actual exception decision if applicable, expected waiting time, and a recipient-benefit assessment against contemporary non-LT care. Do not promise priority or treat DCD, an extended-criteria graft or machine perfusion as evidence that no other candidate could benefit. [1] [6]
- For LDLT, retain independent donor advocacy and donor-team assessment, freedom to withdraw, and separate consent covering uncertain recipient benefit, operative and long-term morbidity, recovery, financial burden and possible retransplantation. Do not relax donor safety criteria or make possession of a living donor a clinical eligibility requirement. [1] [6] [29]
- The living-donor literature reports low but nonzero mortality and incomplete recovery follow-up. Use current center- and graft-specific counseling; do not pool overlapping registry studies or portray donor survival as proof of complete recovery. [29]

ALLOC-B1 — WHAT ARE WE APPROVING?

Evidence and uncertainty before the shared rule

The separation is a governance requirement, not an empirical survival cutoff. A clinically appropriate patient can lack exception eligibility, and an approved exception does not prove clinical benefit. LDLT requires an independent donor determination; a registry entry is not permission to perform experimental transplantation. [1] [2] [30] [31]

Evidence-to-threshold distinction: Six domains: local checklist architecture, not six mandatory national approvals.; 100% applicable completion: governance safeguard, not an outcome-derived percentage.; Hold pending required authorization: operational implementation of separate clinical, donor and research responsibilities.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require 100% completion of the applicable clinical, allocation, donor and research determinations before activation. [2] [1] [30] [31]

- Complete six separate dated determinations: clinical suitability; standardized-exception eligibility; nonstandard NLRB review status; calculated-MELD access; LDLT donor clearance; and research authorization.
  - Mark nonapplicable domains explicitly with a reason—do not manufacture six approvals for every patient.
- Any required missing, pending or adverse determination is a HOLD.
  - An exception application is not approval, a donor is not recipient benefit, and clinical approval is not study activation.
- Recheck the current policy/guidance version before each request; record actual decision and priority rather than predicted entitlement.

Why retain or change it: Retain 100% completion of applicable determinations with dated responsible owners and reasoned N/A entries. A required unresolved determination means HOLD, not an assumed refusal or an authorization by default.

INFORMATION REQUIRED BEFORE THE DECISION

Signed suitability record; exact allocation route and policy version; exception outcome; donor-team status; study and institutional activation record.

Clinical versus allocation: Policy 9.4 permits discretionary requests where appropriate; absence of favorable diagnosis-specific guidance is not absence of a review mechanism, and a request does not promise approval.

Roadmap / protocol reconciliation: Use the same clinical-versus-allocation separation in every standard and high-risk pathway.

Meeting record: ALLOC-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B2 — WHAT IS THE RELEVANT ORGAN OPPORTUNITY COST?

Evidence and uncertainty before the shared rule

Use pathway-level benefit from an agreed entry date, not recipient-only survival. Compare with contemporary tumor-specific care and actual local access. LDLT avoids one direct deceased-donor allocation but creates healthy-donor risk and may generate rescue graft use. Graft quality/perfusion changes do not make opportunity cost zero. [31] [1] [13] [29] [5] [32] [33]

Evidence-to-threshold distinction: Two reviewers: local independence safeguard, not validated optimum.; 6/12/18 months: sensitivity scenarios for access/dropout, not mandatory waits.; HCC ≥60% five-year ITT OS: one expert trial-design benchmark; not a cross-cancer allocation policy.; All grafts: count initial and rescue/retransplant grafts; donor morbidity remains a separate outcome.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require two independent recipient-benefit reviews and an explicit organ-use case; do not adopt a universal five-year survival cutoff. [32] [29] [31] [1] [13] [5] [33]

- Proposed gate: two reviewers not both responsible for recruitment must document expected incremental benefit over the actual contemporary alternative, with unresolved disagreement referred to the full committee.
- For DDLT, show local expected access/dropout and sensitivity scenarios at 6, 12 and 18 months, total graft use including retransplantation, and plausible alternative recipients.
  - These are planning scenarios, not imposed waiting periods.
- For LDLT, separately show healthy-donor risk and potential downstream deceased-donor graft use.
  - Do not call DCD, extended-criteria grafts or machine perfusion “zero opportunity cost.” No priority changes follow from an AI-selected survival threshold.

Why retain or change it: Retain independent benefit and organ-use reviews, but require disease-specific ITT evidence and explicit uncertainty. No universal five-year survival cutoff or numerical allocation weight is adopted.

INFORMATION REQUIRED BEFORE THE DECISION

Same-eligibility/time-zero comparator; expected access; dropout; donor/graft type; uncertainty intervals; graft loss/retransplant estimate; written independent reviews.

Clinical versus allocation: Only the applicable allocation process determines priority. Favorable post-LT prognosis, evidence grade and urgency are not interchangeable measures.

Roadmap / protocol reconciliation: Compare against the cancer roadmap’s actual resection, ablation, HAI, radiation or systemic alternative—not historical untreated survival.

Meeting record: ALLOC-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B3 — WHAT MUST DONOR CONSENT AND FOLLOW-UP INCLUDE?

Evidence and uncertainty before the shared rule

Donor mortality is low but not zero; complication, recovery and financial burdens are more frequent. Registry missingness and selected healthy donors limit reassurance. Consent must include uncertain recipient benefit,

recurrence and possible early graft loss, without pressure to proceed or relax donor criteria. [29] [34] [35] [36] [1]

Quantitative donor context: Kelly reports 8/8,619 deaths by 90 days across eras; A2ALL recorded ≥1 complication in 296/740 donors. These are different outcomes/eras, not poolable rates. The complete donor evidence and follow-up limitations are in ALLOC-B3.

Evidence-to-threshold distinction: Baseline/6 weeks/6 months/1/2 years then annually to five: proposed support schedule; current mandatory reporting remains separately checked.; One sentinel event: conservative review/pause trigger, not a statistically estimated harm boundary.; No relaxed donor criteria: protection against shifting uncertain recipient benefit onto a healthy person.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Do not relax donor criteria; fund independent follow-up and pause after a serious sentinel donor event. [1] [29] [34] [35] [36]

- Proposed readiness gate: independent donor evaluation/advocacy, documented freedom to withdraw, recipient-benefit uncertainty consent, and funded medical, psychological, recovery and financial follow-up before investigational LDLT.
- Proposed local follow-up: baseline, six weeks, six months, one year, two years, then annually through five years.
  - Meet current OPTN reporting duties separately; this five-year support plan is not claimed to be a national reporting requirement.
- One donor death, life-threatening complication, permanent major disability or liver failure requiring transplantation triggers an immediate pause of the relevant investigational LDLT activity and independent review.
  - Other serious donor harm also warrants review; reopening needs independent authorization.

Why retain or change it: Retain independent advocacy, ability to withdraw, funded medical/psychological/financial follow-up and sentinel-event pause. Add explicit denominator/era caveats to consent estimates and collect local outcomes prospectively.

INFORMATION REQUIRED BEFORE THE DECISION

Graft/resection-specific and center-specific risk; donor recovery/employment baseline; independent consent and advocate records; funded follow-up/support; sentinel-event response owner.

Clinical versus allocation: LDLT clearance is separate from recipient selection; the potential need for retransplantation must be included in organ-use discussions.

Roadmap / protocol reconciliation: The same independent donor safeguards apply whether recipient disease is standard or experimental.

Meeting record: ALLOC-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

6 POST TRANSPLANT IMMUNOSUPPRESSION AND GRAFT MONITORING

- Use the approved institutional induction and maintenance regimen with individualized renal, rejection, infection and drug-interaction review. Do not mandate ATG, tacrolimus 12–15 ng/mL, accelerated steroid withdrawal or routine protocol biopsies solely because an ICI was previously given. [6] [25] [24]

- Monitor liver tests, graft vascular patency, drug exposure and infection according to early LT practice; promptly investigate unexplained graft dysfunction and obtain for-cause biopsy when appropriate. Continue rejection vigilance beyond the first month: later events occur, and an early 7–10-day window is inadequate. [26]
- Consider mTOR-based modification only after appropriate healing and vascular assessment, using the current product label and transplant-pharmacy plan. Do not promise recurrence prevention: the SiLVER trial did not demonstrate a significant long-term primary recurrence-free-survival benefit. [37]
- PD-L1, lymphocyte subsets and other immune markers remain optional research correlates, not validated clearance tests.

7 POST TRANSPLANT ONCOLOGIC SURVEILLANCE

- Proposed local schedule for approval: chest CT and contrast liver CT/MRI with AFP every three months for the first two years, then every six months through year five; subsequent surveillance is individualized. Adjust modality to renal function and MRI/CT suitability. This is an operational proposal, not a proven optimal schedule. [3]
- Use explant risk assessment, including RETREAT where applicable, to inform discussion. RETREAT was not validated as a prospective eligibility rule for treated PVTT. Investigate a sustained AFP rise or suspicious lesion; ctDNA alone neither diagnoses recurrence nor authorizes treatment. [3]

8 MANAGEMENT OF RECURRENCE

- Confirm recurrence and restage in the transplant-oncology MDT. Consider resection, ablation or radiation for selected limited disease and systemic therapy for disseminated disease, accounting for graft function, drug interactions and patient priorities. [3]
- Post-LT ICI administration is a different risk setting from pre-LT exposure. It is not routine care and may cause rejection, graft loss and death. Any exceptional use requires a documented transplant-oncology risk assessment and monitoring plan; do not quote pre-LT rejection rates as its risk estimate. [38]
- Individualize immunosuppression reduction with hepatology; do not sacrifice graft safety for an unproven anticancer effect.

9 DATA COLLECTION AND SAFETY OVERSIGHT

- Register every referred or screened candidate in the institutionally approved system, not only recipients. Record acceptance or rejection, treatment entry, response, listing, exception decision, conversion to resection or ablation, progression, toxicity, decompensation, death, patient withdrawal and access-related loss.
- Report overall survival from the prespecified entry point and separately from listing and transplantation. Keep recurrence-free survival, recurrence incidence, graft survival and survival after recurrence distinct, with definitions, confidence intervals, numbers at risk and follow-up completeness. Do not call recipient-only survival an intention-to-treat outcome.
- Capture 90-day recipient morbidity and mortality, biopsy-defined rejection, infection, arterial and biliary complications, graft loss, retransplantation and total grafts used. Record recurrence site, salvage treatment, time off systemic therapy, functional recovery and quality of life, not survival alone.
- Before a transplanting study opens, biostatistics and the independent safety board must approve indication-specific monitoring, evaluable denominators, pause authority and reopening criteria. ALLOC-

B4 now proposes explicit initial-cohort review triggers for ratification; they are not validated efficacy/futility boundaries. No universal cross-cancer five-year survival cutoff is adopted.

- Use a contemporary comparator with the same eligibility and time zero. Account for the interval needed to survive to transplantation, crossover, competing risks and overlapping study cohorts. A nonsignificant difference is not proof of equivalence.

ALLOC-B4 — WHAT DATA ARE REQUIRED BEFORE EXPANSION?

Evidence and uncertainty before the shared rule

A small pilot can test feasibility, staging fidelity, data capture and acute safety, not prove durable survival benefit. Define time zero, all-screened enrollment, waitlist failures, post-LT survival/recurrence, competing death, all grafts, donor harms and patient-reported outcomes before opening. An independent board needs actual pause/reopening authority. [13] [39] [30] [29] [40] [41]

Zero observed events among ten fully followed independent patients still has a one-sided exact 95% binomial upper bound of 25.9% (1−0.05^(1/10)). Thus ten survivors cannot establish a <5% event risk. Ninety percent data completeness permits one missing patient; safety events and vital status require active ascertainment, not tolerance of unaccounted loss.

Evidence-to-threshold distinction: Ten per stratum and interim after five at 90 days: local feasibility cap and safety review cadence.; Twelve-month minimum: earliest expansion discussion, not proof of long-term cancer benefit.; ≥90%: proposed non-vital key-measure completeness threshold; not permission to ignore deaths/graft events or unknown status.; One early death/graft loss/re-LT: review/pause trigger, not automatic futility or withholding rescue.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use a small staged pilot with all-screened accounting and explicit safety pauses; no automatic expansion on favorable recipient survival. [13] [39] [30] [29] [40] [41]

- Proposed initial cap: ten transplants per separately approved biological stratum.
  - Review after the first five have 90-day follow-up; no expansion decision until all first ten have at least 12-month follow-up.
  - Reconcile vital status, graft events and known serious harms for every participant; require ≥90% completeness for other prespecified key outcomes.
- In an initial experimental cohort, one all-cause recipient death, graft loss or retransplantation within 90 days triggers a pause of further experimental enrollment/transplantation pending independent review.
  - A later possibly treatment-related death/graft loss or a donor sentinel event also triggers review/pause.
  - Rescue care is never withheld; reopening requires independent authorization.
- These are proposed conservative review triggers, not statistically validated efficacy/futility boundaries.
  - Before opening, biostatistics and the independent board must ratify or replace them with indication-specific rules, denominators, pause authority and reopening criteria.
- Capture every screened candidate, including nonrecipients, conversion to resection, dropout, deaths and access barriers; report survival from entry and from LT, recurrence, graft use, donor recovery and quality of life.
  - Ten survivors alone do not establish benefit or authorize expansion.

Why retain or change it: Retain the small staged pilot as an OPEN governance proposal, not an efficacy claim. Clarify that known safety/vital-status data must be reconciled for every enrolled patient; ≥90% applies only to other prespecified key measures. Indication-specific statistical design may replace the proposed numbers before opening.

INFORMATION REQUIRED BEFORE THE DECISION

Locked eligibility strata; prospective entry/time-zero dataset; modern comparator; adverse-event definitions; follow-up-completeness report; independently signed statistical and safety plan.

Clinical versus allocation: Study monitoring never changes exception criteria or guarantees future access. Audit access by referral, geography, resources and donor availability without turning protected characteristics into exclusion rules.

Roadmap / protocol reconciliation: Retain cancer-roadmap treatment alternatives and document why non-LT care was chosen when candidates do not proceed.

Meeting record: ALLOC-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

10 REMAINING DECISIONS BEFORE EXPANSION

HCC-B4 — CAN A SURVIVAL-BENEFIT SCORE INFLUENCE ORGAN PRIORITY?

QUESTION AND COMPARISON

HCC candidates under existing US allocation pathways, including whether results can transport to expanded-risk or LDLT cohorts; compare model-informed eligibility/priority with current policy. Evaluate calibration, causal estimand, treatment timing, clinical utility, equity and prospective impact—not only discrimination or correlation.

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Liu 2026 [42] SRTR 2002–2022 HCC cohort restricted to Milan: 30,565 candidates, randomly split into development 15,282 and internal validation 15,283. Table 1 records 19,997 LT and 8,249 dropout events. Primary source | Full-text PDF Locator: Complete local PDF pp2–5, §§1.3, 1.6, 1.8 and 2.3, Table 1. Review depth: Complete local PDF; question-relevant methods/results or guidance sections reviewed.	**Reported result:** Proposed score: 0.65×(C-MELD−6)+1.99×(TBS−2.25)+0.68×log2(AFP). Reported internal correlation .93, R^2 .87 and C-index .91. **Do not infer:** Internal split is not external validation. Abstract refers to MELD 3.0 while Methods §1.6 gives an original-MELD formulation; authors/code clarification is needed. Listing versus post-LT time origins, benefit units and model implementation require reproducibility audit. These are audit questions, not proof of a specific bias. No validated transport to PVTT, LDLT or other cancers.

Study • population • source	Results • interpretation boundary
Liu 2025 [43] SRTR HCC-exception candidates 2002–2019, follow-up to December 2023: 36,634 candidates; 24,457 transplanted, 9,582 dropout. Primary source | Full-text PDF Locator: Complete local 13-page PDF, pp4–6, cohort/Table 1 and statistical methods. Review depth: Complete local 13-page publisher PDF; time-dependent treatment model and units checked.	**Reported result:** Five-year unadjusted restricted-mean survival benefit was 15.7/22.1/25.0 months across increasing TBS groups. Adjusted five-year survival-probability differences were 21.9/34.5/39.4 percentage points. These are different measures, not interchangeable units. **Do not infer:** Overlapping SRTR population is not independent validation of the 2026 score. Methods explicitly report time-dependent LT modeling; do not falsely accuse the study of ignoring treatment timing. Counterfactual assumptions and differing clocks still require scrutiny.
Lai 2017 [44] European ten-center HCC listing cohort, 1987–2015: 2,103 listed, 1,857 transplanted; 636 outside Milan at listing. Primary source | Full-text PDF Locator: New complete 10-page PDF, Methods pp1911–1913, especially p1912 (PDF p3), outcome tables. Review depth: New complete 10-page accepted/full manuscript from university repository; Methods and counterfactual modeling checked.	**Reported result:** The model classified 405/2,103 (19.2%) as no-benefit and 108/2,103 (5.1%) as large-benefit, the latter with a modeled 60-month difference. **Do not infer:** Non-LT counterfactual survival was imputed using MESIAH at the final LT/dropout assessment, not observed in a comparable randomized untreated group. Historical treatment eras and modeling assumptions limit transport. The paper's proposal to delist the no-benefit group is not independently validated clinical utility.
Goudsmit 2023 [45] UNOS 2010–April 2019, adult first-LT population; recipient table: 24,503 non-HCC, 6,922 T2 and 5,448 initially outside criteria. Primary source | TMC: retrieve PDF Locator: Primary full HTML, Methods and Tables 1–2, PMC10 685 016; PDF download unavailable. Review depth: Primary full HTML Methods/Results and Tables 1–2 reviewed; complete PDF not saved.	**Reported result:** Estimated mean benefit over five years from LT was 2.30 years for non-HCC, 1.19 for T2 and 1.45 for initially outside-criteria HCC. **Do not infer:** The calculator is explicitly for patients who actually underwent transplant and met study criteria; it does not directly validate prioritizing every waiting candidate. No external test in PVTT or high-risk LDLT. Do not equate an average-on-treated estimand with conditional benefit for the whole waiting list.

Study • population • source	Results • interpretation boundary
Allen 2024 [33] UK national liver-offering policy evaluation: separate simulation dataset of 1,287 registrations/629 donors plus early real-world NLOS implementation, 2018–2020. Primary source | TMC: retrieve PDF Locator: Primary abstract and full publisher Methods/Results, Table 3, DOI 10.1016/j.jhep.2024.03.020; PDF pending. Review depth: Primary abstract and publisher Methods/Results/Table 3 reviewed; complete PDF not saved.	**Reported result:** Simulation compared observed 2013 deaths/removals 93 (7%) and 4,581 patient-years with modeled TBS outcomes 48 (4%) and 5,262 patient-years. During the first two implementation years, 68% of DBD livers were allocated by TBS; observed waitlist mortality decreased (P<.0001), postlisting survival improved (P=.005), with no adverse post-LT survival signal. **Do not infer:** Simulated gains are not observed outcomes. Before/after implementation is nonrandomized and can reflect secular changes. Nonetheless it is meaningful evidence that a governed benefit-based system can be deployed, unlike a claim that all such models are unusable.
Prosepe 2026 [46] Methods study with simulation and Eurotransplant cirrhosis cohort, 2007–2019: 9,868 candidates; 4,998 LT, 3,013 deaths before LT, 1,361 deaths within three years after LT. Primary source | Full-text PDF Locator: Complete local 15-page PDF, §§4–5, pp9–12; DOI 10.1177/0 962 280 226 1 420 699. Review depth: Complete local 15-page publisher early-online PDF; conditional-benefit assumptions and application checked.	**Reported result:** The in-sample allocation exercise estimated 0.46 additional life-years per graft versus MELD allocation. **Do not infer:** Modeled, not observed implementation benefit; not HCC-specific validation. Identification requires consistency, exchangeability, positivity and correct models. The whole-waitlist conditional estimand differs from average benefit among actual recipients.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Benefit modeling is clinically important because acceptable post-LT survival alone ignores deaths or progression without transplantation. However, a prognostic model, a causal benefit estimate and an allocation policy are different products. High internal discrimination or correlation cannot establish calibration, appropriate treatment-time handling, fairness or improved decisions. The existing score should therefore be studied, not used to override eligibility or priority. The UK experience provides a constructive precedent for eventual governed implementation after population-specific validation and policy integration. [42] [43] [44] [45] [33] [46]

- The UK implementation and longitudinal causal methods show that benefit-based allocation need not remain hypothetical. Favorable internal performance and the larger-benefit signal at higher tumor burden justify further evaluation, not automatic adoption. Conversely, an imputed nontransplant counterfactual, recipient-only scope or overlapping registry sample cannot serve as independent validation. The proper critique is specific to estimand, calibration and transport—not a blanket claim that observational modeling is invalid or that time-dependent LT was never considered.

Certainty / applicability: Moderate that benefit-based allocation is feasible within an established national governance system; low that any particular observational score improves individual HCC decisions; very low

for deployment of the 2026 Liu score in expanded-risk/PVTT/LDLT patients. No independent prospective impact or external high-risk calibration study of that exact score was identified in this bounded search.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Zero current decision weight is a proposed governance rule, not a published numerical efficacy threshold. No arbitrary calibration/C-index cutoff is endorsed here. Acceptance criteria should be prespecified with biostatistics and allocation stakeholders using clinically meaningful calibration error, decision consequences and equity, then independently tested. Findings in within-Milan DDLT cannot simply be extrapolated to PVTT or LDLT.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Liu 2026 [42] "developed and internally validated" Published PDF p2, introduction Source | Full-text PDF	A split-sample internal assessment is not independent external validation.
Liu 2026 [42] "Recipients of living donor liver transplantation (LDLT) were excluded" Published PDF p2, section 1.3 Source | Full-text PDF	LDLT decisions are outside this model's evaluated population.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Give the survival-benefit score zero clinical or allocation weight now; evaluate it in shadow mode only. [42] [43] [44] [45] [33] [46]

- Proposed decision threshold: no score-driven listing, exclusion, exception priority, organ acceptance, LDLT selection or treatment change before external validation and prospective impact review.
- Require a locked reproducible formula and data dictionary, clarification of the MELD/AFP definitions and time zero, independent external calibration and utility testing against current care, equity assessment, and appropriate policy authorization before deployment.
- Do not extrapolate the within-Milan, non-LDLT model to PVTT, LDLT or other cancers.
    - No numerical score cutoff is proposed because its clinical use is not yet justified.

Why retain or change it: OPEN Analyst A position: assign the new score zero clinical eligibility or allocation weight; permit shadow-mode research only. First resolve and lock formula, MELD version, AFP transformation, input dates, predicted horizon and units. Then test independent temporal/geographic external calibration, treatment-time and donor assumptions, net benefit versus current practice, and subgroup equity. Prespecify missing-data handling and failure rules, audit code, and obtain institutional/allocation-policy authorization before any decision support. No score may silently replace OPTN/NLRB policy.

INFORMATION REQUIRED BEFORE THE DECISION

Reproducible model code; independent cohort; time-dependent transplant analysis; competing-event definitions; calibration, utility and subgroup audit.

Clinical versus allocation: Neither a model estimate nor an internal committee can independently rewrite national allocation priority.

Roadmap / protocol reconciliation: Use the score only as a research output; the current clinical roadmap remains the decision authority.

Meeting record: HCC-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

The group must record its final wording, owner, date and approvals for HCC-B1–B4 and ALLOC-B1–B4. This edition implements evidence corrections and makes the unresolved choices visible; it does not record a committee decision that has not occurred.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

REFERENCES

[1] Organ Procurement and Transplantation Network. OPTN Policies: liver allocation and living donation. HRSA/OPTN. 2026. Official source | Full text PDF

Policy or professional guidance, not a primary treatment study. Official full PDF checked.

[2] Organ Procurement and Transplantation Network. Adult MELD Exceptions for Transplant Oncology. HRSA/OPTN. 2025. Official source | Full text PDF

Allocation review guidance. Complete local primary PDF available; relevant methods, selection, results and limitations reviewed.

[3] Singal, Amit G; Llovet, Josep M; Yarchoan, Mark; et al. AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma. Hepatology (Baltimore, Md.). 2023;78(6):1922-1965. DOI | Full text PDF

Policy or professional guidance, not a primary treatment study. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[4] Xu, Edison; Tabrizian, Parissa; Gutierrez, Julio; et al. Downstaging of hepatocellular carcinoma before liver transplantation: Results from a national multicenter prospective cohort study. Hepatology (Baltimore, Md.). 2025;82(3):612-625. DOI | Full text PDF

Prospective observational cohort. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[5] Mazzaferro, Vincenzo; Citterio, Davide; Bhoori, Sherrie; et al. Liver transplantation in hepatocellular carcinoma after tumour downstaging (XXL): a randomised, controlled, phase 2b/3 trial. The Lancet Oncology. 2020;21(7):947-956. DOI | Full text PDF

Randomized trial after successful downstaging. Complete accepted manuscript including protocol appendix plus final published abstract; version-dependent hazard ratios not merged..

[6] Dove, Lorna; Chadha, Ryan M.; Lai, Jennifer C.; et al. AASLD AST Practice Guideline on adult liver transplantation: Candidate evaluation. Hepatology. 2026;83(6):1609-1645. DOI | Full text PDF

Society practice guideline. Complete 60-page PMC manuscript saved; relevant oncology evaluation and donor sections reviewed.

[7] Yao, Francis Y.; Mehta, Neil; Flemming, Jennifer; et al. Downstaging of hepatocellular cancer before liver transplant: Long-term outcome compared to tumors within Milan criteria. Hepatology. 2015;61(6):1968-1977. DOI | Local PDF pending

Clinical study / guidance. Primary PMC full text, Methods/Results and outcome figures reviewed; complete PDF not saved..

[8] Mehta, Neil; Frenette, Catherine; Tabrizian, Parissa; et al. Downstaging Outcomes for Hepatocellular Carcinoma: Results From the Multicenter Evaluation of Reduction in Tumor Size before Liver Transplantation (MERITS-LT) Consortium. Gastroenterology. 2021;161(5):1502-1512. DOI | Full text PDF

Clinical study / guidance. Complete local author manuscript reviewed, including outcome tables..

[9] Sinha, Jasmine; Mehta, Neil; Dodge, Jennifer L.; et al. Are There Upper Limits in Tumor Burden for Down -Staging of Hepatocellular Carcinoma to Liver Transplant? Analysis of the All-Comers Protocol. Hepatology. 2019;70(4):1185-1196. DOI | Local PDF pending

Clinical study / guidance. Published primary abstract only; complete full text needed..

[10] Natarajan, Brahma; Tabrizian, Parissa; Hoteit, Maarouf; et al. Downstaging hepatocellular carcinoma before liver transplantation: A multicenter analysis of the “all-comers” protocol in the Multicenter Evaluation of Reduction in Tumor Size before Liver Transplantation (MERITS-LT) consortium. American Journal of Transplantation. 2023;23(11):1771-1780. DOI | Local PDF pending

Clinical study / guidance. Primary PMC Methods/Results and outcome tables reviewed; complete PDF not saved..

[11] Huang, Annsa C.; Dodge, Jennifer L.; Yao, Francis Y.; et al. National Experience on Waitlist Outcomes for Down-Staging of Hepatocellular Carcinoma: High Dropout Rate in All-Comers. Clinical Gastroenterology and Hepatology. 2023;21(6):1581-1589. DOI | Local PDF pending

Clinical study / guidance. Primary abstract and publisher Methods/Results reviewed; complete PDF not saved..

[12] Tabrizian, Parissa; Holzner, Matthew L.; Mehta, Neil; et al. Ten-Year Outcomes of Liver Transplant and Downstaging for Hepatocellular Carcinoma. JAMA Surgery. 2022;157(9):779-788. DOI | Local PDF pending

Clinical study / guidance. Primary PMC Methods/Results and survival tables reviewed; complete PDF not saved..

[13] Llovet, Josep M; Mauro, Ezequiel; Rimassa, Lorenza; et al. Trial design and end points in hepatocellular carcinoma: an EASL-AASLD-ILCA consensus statement. Nature reviews. Clinical oncology. 2026;23(8):619-647. DOI | Full text PDF

Consensus/research-design guidance, not a primary cohort. Complete local 66-page consensus PDF; high-risk HCC trial-design sections pp19–22. Consensus benchmark, not primary validation..

[14] Yoon, Young-In; Moon, Deok-Bog; Lee, Sung-Gyu; et al. Living-donor liver transplantation in hepatocellular carcinoma with portal-vein tumor thrombosis. International Journal of Surgery. 2026;112(3):7259-7270. DOI | Full text PDF

Retrospective cohort. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[15] Choi, Ho Joong; Kim, Dong Goo; Na, Gun Hyung; et al. The clinical outcomes of patients with portal vein tumor thrombi after living donor liver transplantation. Liver Transplantation. 2017;23(8):1023-1031. DOI | Local PDF pending

Retrospective selected-recipient LDLT cohort. Primary published abstract only; complete full text needed..

[16] Yang, Z; Luo, FZ; Wang, S; et al. Alpha-fetoprotein and 18F-FDG standard uptake value predict tumor recurrence after liver transplantation for hepatocellular carcinoma with portal vein tumor thrombosis: Preliminary experience. Hepatobiliary & Pancreatic Diseases International. 2020;19(3):229-234. DOI | Local PDF pending

Retrospective selected-recipient LT cohort. Primary journal abstract only; complete full text needed..

[17] Ye, Zhaodan; Zhuang, Li; Liu, Xiangyan; et al. The clinical outcomes of patients with vascular invasion after deceased donor liver transplantation. Journal of Gastrointestinal Oncology. 2024;15(4). DOI | Full text PDF

Retrospective selected-recipient DDLT cohort. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[18] Soin, Arvinder S; Bhangui, Prashant; Kataria, Tejinder; et al. Experience With LDLT in Patients With Hepatocellular Carcinoma and Portal Vein Tumor Thrombosis Postdownstaging. Transplantation. 2020;104(11):2334-2345. DOI | Local PDF pending

Selected-recipient cohort following a prospective downstaging protocol. Primary abstract and publisher HTML Methods/Results reviewed; complete PDF not saved..

[19] Assalino, Michela; Terraz, Sylvain; Grat, Michal; et al. Liver transplantation for hepatocellular carcinoma after successful treatment of macrovascular invasion – a multi-center retrospective cohort study. Transplant International. 2020;33(5):567-575. DOI | Full text PDF

Retrospective selected-recipient cohort. New complete 9-page publisher PDF, including Methods, anatomy definitions, results, tables and references..

[20] Sposito, Carlo; Berardi, Giammauro; Viganò, Mauro; et al. Outcomes of Y90-radioembolization as downstaging to liver transplantation in patients with hepatocellular carcinoma and tumoral portal vein thrombosis. Hepatology. 2026. DOI | Local PDF pending

Clinical study / guidance. Published primary abstract only; complete full text needed..

[21] Tabrizian, Parissa; Holzner, Matthew L.; Ajmera, Veeral; et al. Intention-to-treat outcomes of patients with hepatocellular carcinoma receiving immunotherapy before liver transplant: The multicenter VITALITY study. Journal of Hepatology. 2025;82(3):512-522. DOI | Local PDF pending

Multicenter observational cohort. Primary published abstract and accessible publisher material; complete full text needed..

[22] Bhoori, Sherrie; Rivoltini, Licia; Pinato, David J.; et al. Efficacy of liver transplantation after response to atezolizumab-bevacizumab downstaging of intermediate and advanced hepatocellular carcinoma (ImmunoXXL). Journal of Hepatology. 2026;85(1):117-129. DOI | Local PDF pending

Prospective single-center phase II study. Primary abstract and accessible publisher Results sections; complete full text needed..

[23] Moeckli, Beat; Wassmer, Charles-Henri; El Hajji, Sofia; et al. Determining safe washout period for immune checkpoint inhibitors prior to liver transplantation: An international retrospective cohort study. Hepatology. 2025;82(5):1122-1137. DOI | Local PDF pending

Retrospective international cohort. Primary published abstract; complete full text needed. Repository returned access challenge..

[24] Rezaee-Zavareh, Mohammad Saeid; Yeo, Yee Hui; Wang, Tielong; et al. Impact of pre-transplant immune checkpoint inhibitor use on post-transplant outcomes in HCC: A systematic review and individual patient data meta-analysis. Journal of Hepatology. 2025;82(1):107-119. DOI | Full text PDF

Secondary systematic review/individual-patient-data meta-analysis, not a new primary trial. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[25] Fang, Jun; Zhong, Siyi; Wang, Tielong; et al. Immune-related adverse events are a potent predictor of post-transplant rejection in HCC: a multicentre retrospective cohort study. Gut. 2026;75(6):1201-1210. DOI | Full text PDF

Retrospective cohort. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[26] Aceituno, L.; Magyar, C.; Tabrizian, P.; et al. Safety of Immune Checkpoint Inhibitors Prior to Liver Transplantation in Hepatocellular Carcinoma. Alimentary Pharmacology & Therapeutics. 2026;63(10):1369-1379. DOI | Full text PDF

Retrospective matched cohort. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[27] Bauer, Ulrike; Troppmair, Maria R; Khaled, Najib Ben; et al. Liver Transplantation After Immune Checkpoint Inhibition in Hepatocellular Carcinoma-Data From the Multinational LITCHI Registry. Liver international : official journal of the International Association for the Study of the Liver. 2026;46(6):e70 692. DOI | Local PDF pending

Retrospective multicenter registry. Primary full HTML and indexed publisher PDF Methods/Results reviewed; actual complete PDF download blocked..

[28] Ong, Meiching; He, Kang; Wang, Feng; et al. Immune Checkpoint Inhibitor–Based Downstaging Therapy for Hepatocellular Carcinoma. JAMA Oncology. 2026. DOI | Local PDF pending

Retrospective matched cohort. Primary published abstract; complete full text needed..

[29] Kelly, Morgan E.; Truby, Tianna M.; Lai, Jennifer C.; et al. Three Decades of Living Liver Donor Outcomes: A National Cohort Analysis of Patterns in Mortality and Recovery. Journal of Transplantation. 2026;2026(1):2 545 887. DOI | Full text PDF

National retrospective donor cohort. Original full PDF checked.

[30] US Department of Health and Human Services. 45 CFR 46.111: Criteria for IRB approval of research. . 2018. Official source | Local PDF pending

Clinical research / guidance. Official regulation checked online; institutional applicability must be determined.

[31] OPTN Ethics Committee. Ethical Principles in the Allocation of Human Organs. . 2015. Official source | Local PDF pending

Clinical research / guidance. Official full HTML reviewed; ethics white paper, not allocation policy.

[32] Adam, René; Piedvache, Céline; Chiche, Laurence; et al. Liver transplantation plus chemotherapy versus chemotherapy alone in patients with permanently unresectable colorectal liver metastases (TransMet): results from a multicentre, open-label, prospective, randomised controlled trial. Lancet (London, England). 2024;404(10 458):1107-1118. DOI | Full text PDF

Multicenter randomized trial. Complete local primary PDF available; relevant methods, selection, results and limitations reviewed.

[33] Allen, Elisa; Taylor, Rhiannon; Gimson, Alexander; et al. Transplant benefit-based offering of deceased donor livers in the United Kingdom. Journal of Hepatology. 2024;81(3):471-478. DOI | Local PDF pending

Primary policy simulation and before-after implementation evaluation. Primary abstract and publisher Methods/Results/Table 3 reviewed; complete PDF not saved..

[34] Abecassis, M.M.; Fisher, R.A.; Olthoff, K.M.; et al. Complications of Living Donor Hepatic Lobectomy—A Comprehensive Report. American Journal of Transplantation. 2012;12(5):1208-1217. DOI | Local PDF pending

Clinical research / guidance. Primary full HTML Methods, Results and complication tables reviewed; local PDF pending.

[35] DiMartini, A.; Dew, M.A.; Liu, Q.; et al. Social and Financial Outcomes of Living Liver Donation: A Prospective Investigation Within the Adult-to-Adult Living Donor Liver Transplantation Cohort Study 2 (A2ALL-2). American Journal of Transplantation. 2017;17(4):1081-1096. DOI | Local PDF pending

Clinical research / guidance. Primary full HTML financial-results section and abstract verified; local PDF pending.

[36] Xiao, Jieling; Zeng, Rebecca Wenling; Lim, Wen Hui; et al. The incidence of adverse outcome in donors after living donor liver transplantation: A meta-analysis of 60,829 donors. Liver Transplantation. 2024;30(5):493-504. DOI | Local PDF pending

Clinical research / guidance. Primary journal abstract verified; complete PDF and overlap audit pending.

[37] Geissler, Edward K; Schnitzbauer, Andreas A; Zülke, Carl; et al. Sirolimus Use in Liver Transplant Recipients With Hepatocellular Carcinoma: A Randomized, Multicenter, Open-Label Phase 3 Trial. Transplantation. 2016;100(1):116-25. DOI | Full text PDF

Randomized immunosuppression trial. Local full PDF reviewed.

[38] De Martin, Eleonora; Antonini, Teresa Maria; Uguen, Thomas; et al. Immune checkpoint inhibitors after liver transplantation: The role of immunosuppressive management. American Journal of Transplantation. 2026;26(8):1958-1971. DOI | Local PDF pending

Retrospective multicenter cohort. Published full article; primary abstract verified, full text needed.

[39] Eldridge, Sandra M; Chan, Claire L; Campbell, Michael J; et al. CONSORT 2010 statement: extension to randomised pilot and feasibility trials. BMJ. 2016;:i5239. DOI | Local PDF pending

Clinical research / guidance. Primary journal HTML checklist, scope and explanations reviewed; complete PDF pending.

[40] Maspero, Marianna; Sposito, Carlo; Bongini, Marco A; et al. Liver Transplantation for Intrahepatic Cholangiocarcinoma After Chemotherapy and Radioembolization: An Intention-To-Treat Study. Transplant international : official journal of the European Society for Organ Transplantation. 2024;37:13 641. DOI | Full text PDF

Single-center intent-to-treat protocol cohort. Full text reviewed.

[41] Yaqub, Sheraz; Busund, Sondre; Smedman, Tor Magnus; et al. Liver transplantation for locally advanced non-resectable intrahepatic cholangiocarcinoma treated with neoadjuvant therapy: early results from the TESLA trial. British Journal of Surgery. 2025. DOI | Local PDF pending

Clinical study / guidance. Complete indexed primary research-letter text and Table 1/Figure 1 read; supplementary eligibility file not independently reviewed; PDF download failed..

[42] Liu, Hao; Neckermann, Isabel; Mial-Anthony, Jason; et al. Optimizing Liver Transplant Allocation for Hepatocellular Carcinoma: Development and Validation of a Survival Benefit-Based Model. Clinical Transplantation. 2026;40(3):e70 488. DOI | Full text PDF

Retrospective prediction model. Complete local PDF; question-relevant methods/results or guidance sections reviewed..

[43] Liu, Hao; Zhang, Wei; Di, Mengyang; et al. Survival benefit associated with liver transplantation for hepatocellular carcinoma based on tumor burden scores at listing. Hepatology Communications. 2025;9(1):e0619. DOI | Full text PDF

Clinical study / guidance. Complete local 13-page publisher PDF; time-dependent treatment model and units checked..

[44] Lai, Quirino; Vitale, Alessandro; Iesari, Samuele; et al. Intention-to-Treat Survival Benefit of Liver Transplantation in Patients With Hepatocellular Cancer. Hepatology. 2017;66(6):1910-1919. DOI | Full text PDF

Clinical study / guidance. New complete 10-page accepted/full manuscript from university repository; Methods and counterfactual modeling checked..

[45] Goudsmit, Ben F.J.; Ilaria Prosepe, ; Tushuizen, Maarten E.; et al. Survival benefit from liver transplantation for patients with and without hepatocellular carcinoma. JHEP Reports. 2023;5(12):100 907. DOI | Local PDF pending

Clinical study / guidance. Primary full HTML Methods/Results and Tables 1–2 reviewed; complete PDF not saved..

[46] Prosepe, Ilaria; van Geloven, Nan; de Ferrante, Hans; et al. Estimating conditional survival benefit for the allocation of scarce resources. Statistical Methods in Medical Research. 2026;35(4):812-826. DOI | Full text PDF

Primary methods/simulation study with registry application. Complete local 15-page publisher early-online PDF; conditional-benefit assumptions and application checked..

REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

Problem: The earlier B proposals had an uneven evidence base and emphasized a stance before its supporting and contrary data. **Correction and reason:** Added question-level primary-study tables with selection, denominators, outcome clocks, harms, limitations, exact source excerpts and threshold provenance before each OPEN proposal. Sources/PDF links are adjacent; missing full texts remain flagged.

Problem: Local operational numbers could be mistaken for validated eligibility or allocation thresholds. **Correction and reason:** Separated outcome-derived estimates, guideline/policy wording and local safeguards; revised proposals when the deeper review changed their evidentiary basis. Institutional decisions remain blank.

Approval record Clinical lead _____ Allocation review _____ Research authorization if required _____ Donor review if applicable _____ Date _____